LAPORAN WORKSHOP DEVELOPER OPERATIONAL

MINGGU KE- 9

Materi Pembelajaran :

Mengenal dan Konfigurasi Docker Swarm

oleh

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PROGRAM STUDI TEKNIK INFORMATIKA

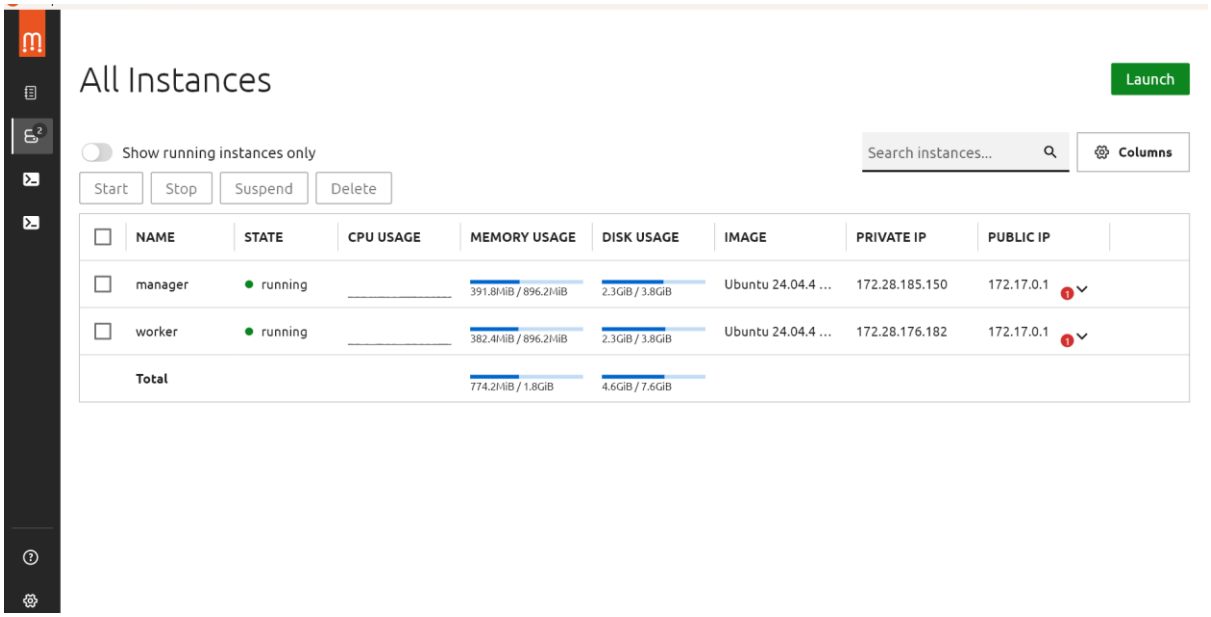
JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

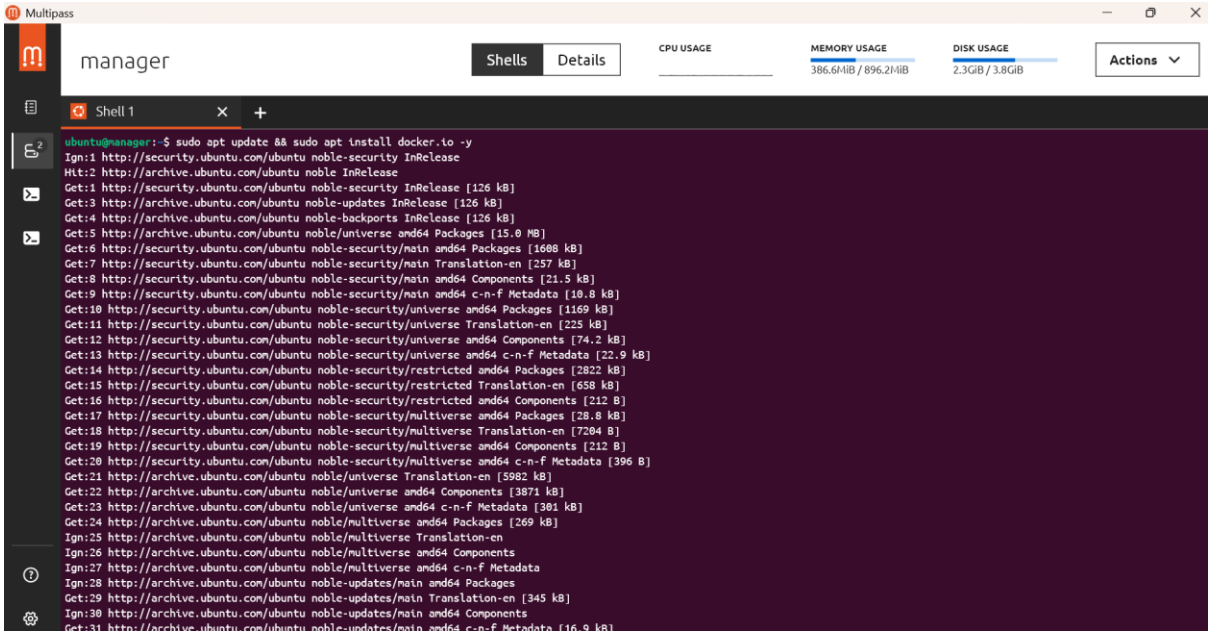
2025

Acara 17 dan Acara 18

1. Tampilan Manager dan Worker



2. Install Docker



3. Mengambil token swam

```
ubuntu@manager:~$ sudo docker swarm init --advertise-addr 172.28.185.150
Swarm initialized: current node (uotci1ukt0aytkqqxyuueg6fo) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-1kttulb8km02f7kgvlbk3atvsbnwgqkkhxmd0u361lqbm5slr8-2w1hcgr6vjco1apummvmwsmlh 172.28.185.150:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

4. Worker join melalui token yang telah diseakan oleh manager

```
ubuntu@worker:~$ sudo docker swarm join --token SWMTKN-1-1kttulb8km02f7kgvlbk3atvsbnwgqkkhxmd0u361lqbm5slr8-2w1hcgr6vjco1apummvmwsmlh 172.28.185.150:2377
This node joined a swarm as a worker.
```

5. **Install nginx dengan port 8080**

```
ubuntu@manager:~$ sudo docker service create --name web-tugas --publish 80:80 --replicas 3 nginx:alpine
ricvcobkgkwaso0mg0dmvuhss
overall progress: 3 out of 3 tasks
1/3: running
2/3: running
3/3: running
verify: Service ricvcobkgkwaso0mg0dmvuhss converged
```

6. **Berikutnya periksalah node yang terhubung dalam docker swarm dengan menjalankan perintah docker node ls.**

```
ubuntu@manager:~$ sudo docker info |grep -i swarm
 Swarm: active
```

7. **Apabila ingin melihat kembali informasi token untuk join sebagai worker, maka jalankanlah perintah docker swarm join-token worker. Untuk menambahkan manager dapat menggunakan token dengan menjalankan perintah docker swarm join-token manager.**

```
ubuntu@manager:~$ sudo docker service ls
ID             NAME        MODE         REPLICAS   IMAGE          PORTS
ricvcobkgkwa   web-tugas   replicated   5/5        nginx:alpine   *:80->80/tcp
ubuntu@manager:~$
```

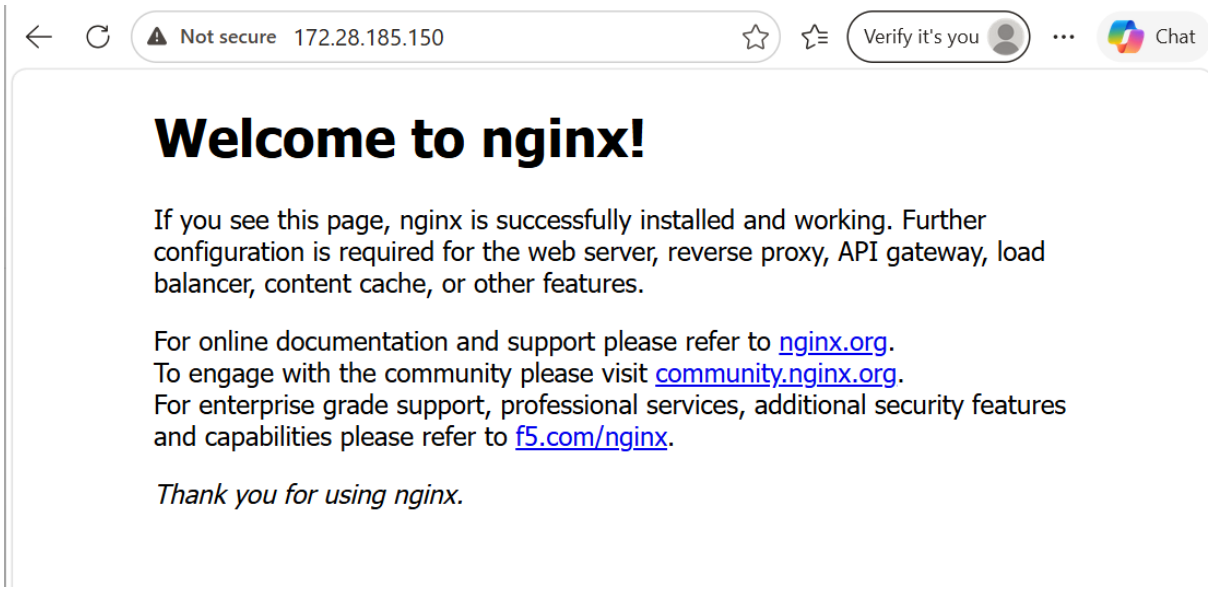
8. **Cek Ip manager lalu mengambil ip yang eth0**

```
ubuntu@manager:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 52:54:00:4c:fb:e1 brd ff:ff:ff:ff:ff:ff
    inet 172.28.185.150/20 metric 100 brd 172.28.191.255 scope global dynamic eth0
       valid_lft 82999sec preferred_lft 82999sec
    inet6 fe80::5054:ff:fe4c:fbe1/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:0d:38:7a:ef brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
7: docker_gwbridge: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group defaul
    link/ether 02:42:6c:77:c3:58 brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global docker_gwbridge
       valid_lft forever preferred_lft forever
    inet6 fe80::42:6cff:fe77:c358/64 scope link
       valid_lft forever preferred_lft forever
```

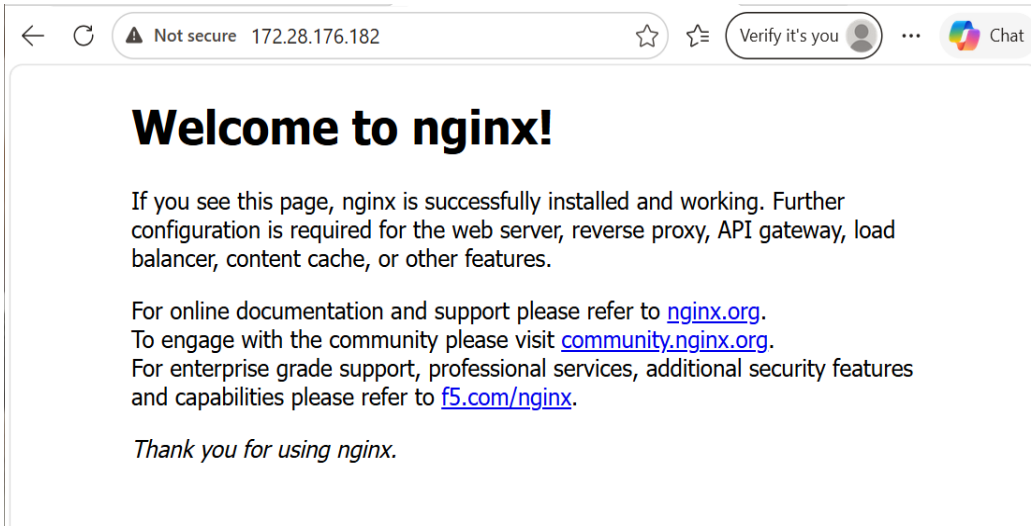
9. **Cek ip worker lalu mengambil ip yang eth0**

```
ubuntu@worker:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 52:54:00:33:52:f0 brd ff:ff:ff:ff:ff:ff
    inet 172.28.176.182/20 metric 100 brd 172.28.191.255 scope global dynamic eth0
       valid_lft 83384sec preferred_lft 83384sec
    inet6 fe80::5054:ff:fe33:52f0/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 72:90:83:82:d2:19 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
7: docker_gwbridge: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 16:6a:b8:c0:73:d0 brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global docker_gwbridge
       valid_lft forever preferred_lft forever
    inet6 fe80::146a:b8ff:fec0:73d0/64 scope link
       valid_lft forever preferred_lft forever
9: vethf609ef8@if8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker_gwbridge st
ate UP group default
    link/ether 3e:d4:24:39:4d:90 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::3cd4:24ff:fe39:4d90/64 scope link
       valid_lft forever preferred_lft forever
```

10. Tampilan nginx pada bagian manager



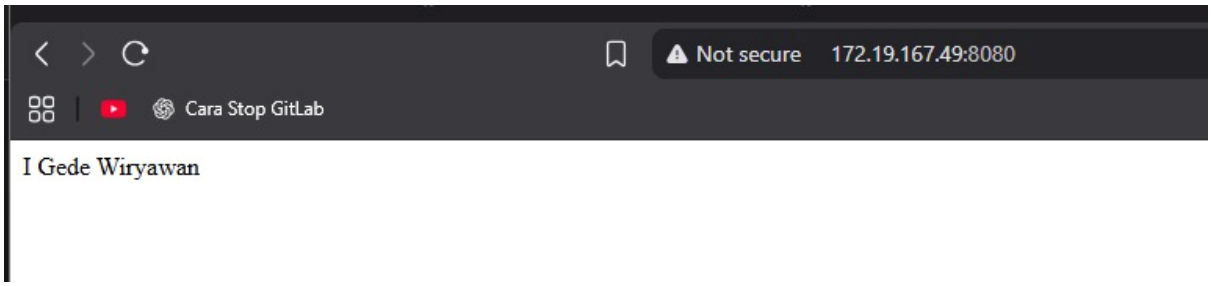
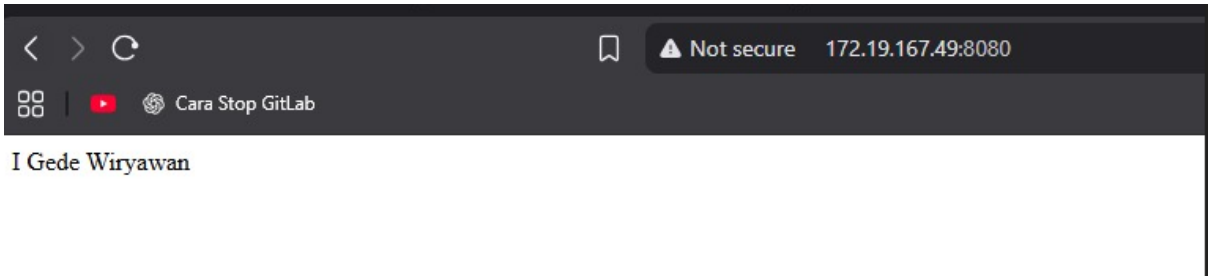
11. Tampilan nginx pada bagian worker



12. Upgrade service image igdw

```
ubuntu@Manajer:~$ sudo docker service update --image igdw/first-repo:latest web
web
overall progress: 3 out of 3 tasks
1/3: running   [==================================================>]
2/3: running   [==================================================>]
3/3: running   [==================================================>]
verify: Service web converged
```

13.Tampilan ip igdw manager



14. Tampilan ip igdw worker